

$$2.8) \quad \dot{x} = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 1 & 2 \\ -2 & 0 & -2 \end{bmatrix} x + \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} u$$

$$A^2 = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 1 & 2 \\ -2 & 0 & -2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ 1 & 1 & 2 \\ -2 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & -3 & -2 \\ -1 & 0 & -1 \\ 0 & 2 & 2 \end{bmatrix}$$

$$C = \begin{bmatrix} B & AB & A^2B \end{bmatrix} = \begin{bmatrix} -2 & -3 & -4 \\ 0 & 0 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

$\text{rank}(C) = 3$ Controllable

Poles: $-7, -1+j2, -1-j2$

$$\begin{aligned} \Phi &= (s+7)(s+1-j2)(s+1+j2) \\ &= (s+7)(s^2+s+2js+s+1+j2-2js-j2+4) \\ &= (s+7)(s^2+2s+5) \\ &= (s^3+2s^2+5s+7s^2+14s+28) = s^3 + \underset{\bar{a}_3}{9}s^2 + \underset{\bar{a}_2}{19}s + \underset{\bar{a}_1}{35} + \underset{\bar{a}_0}{28} \end{aligned}$$

$$\Phi = \det(sI - A)$$

$$\begin{aligned} &= \begin{vmatrix} s-2 & 1 & -1 \\ -1 & (s-1) & -2 \\ 2 & 0 & (s+2) \end{vmatrix} = (s-2)(s-1)(s+2) - (-s-2+4) + \\ &\quad -1(-2s+2) \\ &= (s^2-3s+2)(s+2) + s-2 + 2s-2 \\ &= s^3+2s^2-3s^2-6s+2s+4+3s-4 \\ &= s^3-s^2-s \end{aligned}$$

$$\begin{matrix} 1 & -1 & -1 & 0 \\ \bar{a}_3 & \bar{a}_2 & \bar{a}_1 & \bar{a}_0 + 35 \end{matrix}$$

$$A_c = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad B_c = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$C(A_c, B_c) = [B_c, A_c B_c, A_c^2 B_c] \quad A_c^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$T_c = C C^{-1} \quad C^{-1} = \frac{1}{-1} \begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$T_c = \begin{bmatrix} 2 & -3 & -4 \\ 0 & 0 & 1 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} -1 & -1 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -2 \\ 1 & 0 & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

$$K_c = [-35, -20, -10]$$

$$T_c^{-1} = \frac{1}{-1} \begin{bmatrix} 0 & -1 & 1 \\ -1 & -1 & -0 \\ 0 & -2 & 1 \end{bmatrix}^T = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 1 & 0 \\ 0 & 2 & -1 \end{bmatrix}^T = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 2 \\ -1 & 0 & -1 \end{bmatrix}$$

$$K = [-35, -20, -10] \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 0 & -1 \end{bmatrix} = [-10, -55, -30]$$

$$2.9) \dot{x} = \begin{bmatrix} -9 & 4 & 4 & -5 \\ 6 & -2 & -7 & 9 \\ -5 & 0 & -9 & 5 \\ 0 & -7 & 0 & 4 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \\ -2 \\ 3 \end{bmatrix} u$$

From Matlab:

$$C = [B \ AB \ A^2B \ A^3B] = \begin{bmatrix} 1 & -32 & 528 & -4971 \\ 0 & 47 & -374 & 1611 \\ -2 & 28 & -32 & -3757 \\ 3 & 12 & -281 & 1494 \end{bmatrix}$$

$\text{Rank}(C) = 4 \Rightarrow$ system is controllable

desired poles at $-15, -10, -2+j2, -2-j2$

Char polynomial $\Phi = s^4 + 16s^3 + 96s^2 + 73s + 2499$

$$\Phi = s^4 + 29s^3 + 258s^2 + 800s + 1200$$

$$A_c = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -2499 & -73 & -96 & -16 \end{bmatrix} \quad B_c = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$C_c = [B_c \ A_c B_c \ A_c^2 B_c \ A_c^3 B_c] = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -16 \\ 0 & 1 & -16 & 16 \\ 1 & -16 & 16 & -1097 \end{bmatrix} \quad T_c = C_c^{-1} = \begin{bmatrix} 478 & 112 & 16 & 1 \\ 139 & 378 & 47 & 0 \\ -1727 & 224 & -4 & -2 \\ -1631 & 199 & 60 & 3 \end{bmatrix}$$

$$K_c = [1299, -727, -162, -13]$$

$$K = K_c T_c^{-1} = [-2.1934, -0.5348, 1.2271, -2.7842]$$

$$2.12) a) \quad A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} B & AB \end{bmatrix} \quad \Theta = \begin{bmatrix} C \\ CA \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \quad = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}$$

$$\text{Rank} = 2 \quad \text{Rank} = 1$$

Controllable, not observable.

$$b) \quad \dot{x} = A_x + B_u = A_x + B(-K_x + v) = (A - BK)x + Bv$$

$$X = \left(\begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} K_1 & K_2 \end{bmatrix} \right) = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} K_1 & K_2 \\ K_1 & K_2 \end{bmatrix}$$

$$= \begin{bmatrix} -K_1 & 1-K_2 \\ -K_1 & 2-K_2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} v$$

$$C_c = \begin{bmatrix} 1 & 1-K_1-K_2 \\ 1 & 2-K_1-K_2 \end{bmatrix} \quad \text{controllable for all } K_1, K_2$$

$$\Theta_c = \begin{bmatrix} 0 & 1 \\ -K_1 & 2-K_2 \end{bmatrix} \quad \text{observable for } K_1 \neq 0$$

$$2.13) \dot{X} = \begin{bmatrix} 0 & 10 \\ 1 & 0 \end{bmatrix} X + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad y = \begin{bmatrix} 0 & 1 \end{bmatrix} X$$

$$\Theta = \begin{bmatrix} C \\ CA \end{bmatrix} \quad \Theta = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{rank}(\Theta) = 2$$

poles at -8, -9

$$\bar{\Phi} = (s+8)(s+9) = s^2 + \frac{17}{a_1}s + \frac{72}{a_0}$$

$$A^T = \begin{bmatrix} 0 & 1 \\ 10 & 0 \end{bmatrix} \quad C^T = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$sI - A^T = \begin{bmatrix} s & -1 \\ -10 & s \end{bmatrix} \quad \phi(s) = s^2 - 10$$

$a_0 = -10 \quad a_1 = 0$

$$A_c = \begin{bmatrix} 0 & 1 \\ 10 & 0 \end{bmatrix} \quad B_c = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad K_c = \begin{bmatrix} -10-72 & 0-17 \\ \hline -82 & -17 \end{bmatrix}$$

$$K = K_c T_c^{-1}$$

$$T_c = C_c C_c^{-1} \quad C_c = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad C_c^{-1} = \frac{1}{-1} \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$T_c = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad T_c^{-1} = 1 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$K = \begin{bmatrix} -82 & -17 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -82 & -17 \end{bmatrix}$$

$$G = K^T = \begin{bmatrix} -82 \\ -17 \end{bmatrix}$$

$$\begin{aligned} \hat{X} &= \left(\begin{bmatrix} 0 & 10 \\ 1 & 0 \end{bmatrix} X + \begin{bmatrix} -82 \\ -17 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} \right) X + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u - \begin{bmatrix} -82 \\ -17 \end{bmatrix} y \\ &= \begin{bmatrix} 0 & -72 \\ 1 & -17 \end{bmatrix} X + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u - \begin{bmatrix} -82 \\ -17 \end{bmatrix} y \end{aligned}$$

$$2.14 \quad y = [1 \ 0 \ 0]x \quad \dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & 3 & 5 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

Poles at $-5, -1+j, -1-j$

$$A^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & 3 & 5 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & 3 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & 3 & 5 \\ -10 & 13 & 28 \end{bmatrix} \quad \Theta = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{rank}(\Theta) = 3$$

$$A^T = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 0 & 3 \\ 0 & 1 & 5 \end{bmatrix} \quad C^T = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \Phi(s) = \det(sI - A^T)$$

$$\Phi(s) = \det \begin{bmatrix} s & 0 & 2 \\ -1 & s & -3 \\ 0 & -1 & s-5 \end{bmatrix} = s(s^2 - 5s - 3) - 0 + 2(1-0) = s^3 - 5s^2 - 3s + 2$$

$a_2 \quad a_1 \quad a_0$

$$\begin{aligned} \bar{\Phi}(s) &= (s+5)(s+1-j)(s+1+j) \\ &= (s+5)(s^2 + s + js + s + 1 + j - js - j - j^2) \\ &= (s+5)(s^2 + 2s + 2) = s^3 + 2s^2 + 2s + 5s^2 + 10s + 10 \\ &= s^3 + 7s^2 + 12s + 10 \end{aligned}$$

$\bar{a}_2 \quad \bar{a}_1 \quad \bar{a}_0$

$$A_c^T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & 3 & 5 \end{bmatrix} \quad B_c^T = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad C_c^T = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 5 \\ 1 & 5 & 28 \end{bmatrix} \quad C_c^{-1} = \frac{1}{-1} \begin{bmatrix} 3 & -5 & -1 \\ -5 & -1 & -2 \\ -1 & -2 & 0 \end{bmatrix}^T$$

$$= - \begin{bmatrix} -3 & -5 & -1 \\ -5 & -1 & -2 \\ -1 & -2 & 0 \end{bmatrix}$$

$$28 + 75 - 12 \quad \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & 3 & 5 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 & 1 \\ -2 & 3 & 5 \\ -10 & 13 & 28 \end{bmatrix}$$

$$T_c = C_c \cdot C_c^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -3 & -5 & -1 \\ -5 & -1 & -2 \\ -1 & -2 & 0 \end{bmatrix} = \begin{bmatrix} -3 & -5 & -1 \\ -5 & -1 & -2 \\ -1 & -2 & 0 \end{bmatrix}$$

$$K_c = [2-10, -3-12, -5-7] = [-8, -15, -12]$$

-3-25

$$T_c = \begin{bmatrix} -3 & -5 & 1 \\ -5 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_1} \begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ -3 & -5 & 1 \end{bmatrix}^T$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 5 \\ 1 & 5 & 28 \end{bmatrix} \quad K = K_c T_c^{-1} = [-8, -15, -12] \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 5 \\ 1 & 5 & 28 \end{bmatrix} = [-12, -75, -419]$$

$$-8-75-12 \cdot 28 \quad G = K^T = \begin{bmatrix} -12 \\ -75 \\ -419 \end{bmatrix}$$

$$\dot{\hat{x}} = (A + GC)\hat{x} + Bu - Gy, \quad GC = \begin{bmatrix} -12 \\ -75 \\ -419 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -12 & 0 & 0 \\ -75 & 0 & 0 \\ -419 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -12 & 1 & 0 \\ -75 & 0 & 1 \\ -421 & 3 & 5 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u - \begin{bmatrix} -12 \\ -75 \\ -419 \end{bmatrix} y$$

$$\begin{array}{r} 1 \\ 12 \\ 28 \\ \hline 56 \\ 280 \\ \hline = 336 \end{array}$$

$$\begin{array}{r} 18 \\ 75 \\ 336 \\ \hline 419 \end{array}$$

$$2.15) \dot{x} = \begin{bmatrix} -1 & -5 & 4 \\ -9 & 6 & -5 \\ 0 & 0 & -3 \end{bmatrix} x + \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} u \quad y = \begin{bmatrix} 1 & 2 & 0 \end{bmatrix} x$$

$$P = \begin{bmatrix} -3 & 7 & -129 \\ 0 & 22 & 84 \\ 1 & -3 & 9 \end{bmatrix} \quad \text{rank} = 3 \quad \text{controllable}$$

$$O = \begin{bmatrix} 1 & 2 & 0 \\ -19 & 7 & -6 \\ -44 & 137 & -93 \end{bmatrix} \quad \text{rank} = 3 \quad \text{observable}$$

poles: $-20, -10+j10, -10-j10$

$$\begin{aligned} \bar{\Phi} &= (s+20)(s+10-j10)(s+10+j10) \\ &= (s+20)(s^2 + 10s + 10js + 10s + 100 + 100j - 100j - 100) \\ &= s^3 + 20s^2 + 200s + 20s^2 + 400s + 4000 \\ &= s^3 + 40s^2 + 600s + 4000 \end{aligned}$$

$\bar{a}_3 \quad \bar{a}_2 \quad \bar{a}_1 \quad \bar{a}_0$

$$\det(sI - A) = \det \begin{bmatrix} s+1 & 5 & -4 \\ 9 & s-6 & 5 \\ 0 & 0 & s+3 \end{bmatrix} = (s+1)(s-6)(s+3) - 5(9(s+3))$$

$$= (s^2 - 5s - 6)(s+3) - 45(s+3)$$

$$= s^3 + 3s^2 - 5s^2 - 15s - 6s + 18 - 45s - 135$$

$$= s^3 - 2s^2 - 66s - 117$$

$a_3 \quad a_2 \quad a_1 \quad a_0$

$T_c =$

$$A_c = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 153 & 66 & 2 \end{bmatrix} \quad B_c = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$T_c = O^T C_c^{-1} \quad C_c^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 2 & 70 \end{bmatrix} = \begin{bmatrix} -66 & -2 & 1 \\ -2 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$T_c = \begin{bmatrix} -72 & -21 & 1 \\ -9 & 3 & 2 \\ -81 & -6 & 0 \end{bmatrix}$$

$$K_c = [-153, -4000, -66-600, -2-40]$$

$$= [-4153, -666, -42]$$

$$K = K_c \cdot T_c^{-1}$$

$$= [16.078, -29.039, 40.206]$$

$$\hat{x} = \begin{bmatrix} 16.078 & -29.039 & 40.206 \\ -38.039 & 80.412 & 3 \end{bmatrix} x + \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} u - \begin{bmatrix} 16.078 \\ -29.039 \\ 40.206 \end{bmatrix} y$$